

# Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions





## OVERVIEW

# Dataset at a Glance

3.9K

Total  
Purchases

Transactions analyzed

18

Data Points

Features per  
transaction

\$60

Avg Purchase

Typical transaction  
value

Demographics, purchase details, and shopping behavior across categories, seasons, and customer segments

# Python Analysis Pipeline



Load Data

Clean & Impute

Engineer Features

Standardize

Comprehensive data cleaning including median imputation for 37 missing review ratings, age grouping, and PostgreSQL integration

# Revenue by Gender

## Male Customers

\$157,890 total revenue

68% of total sales

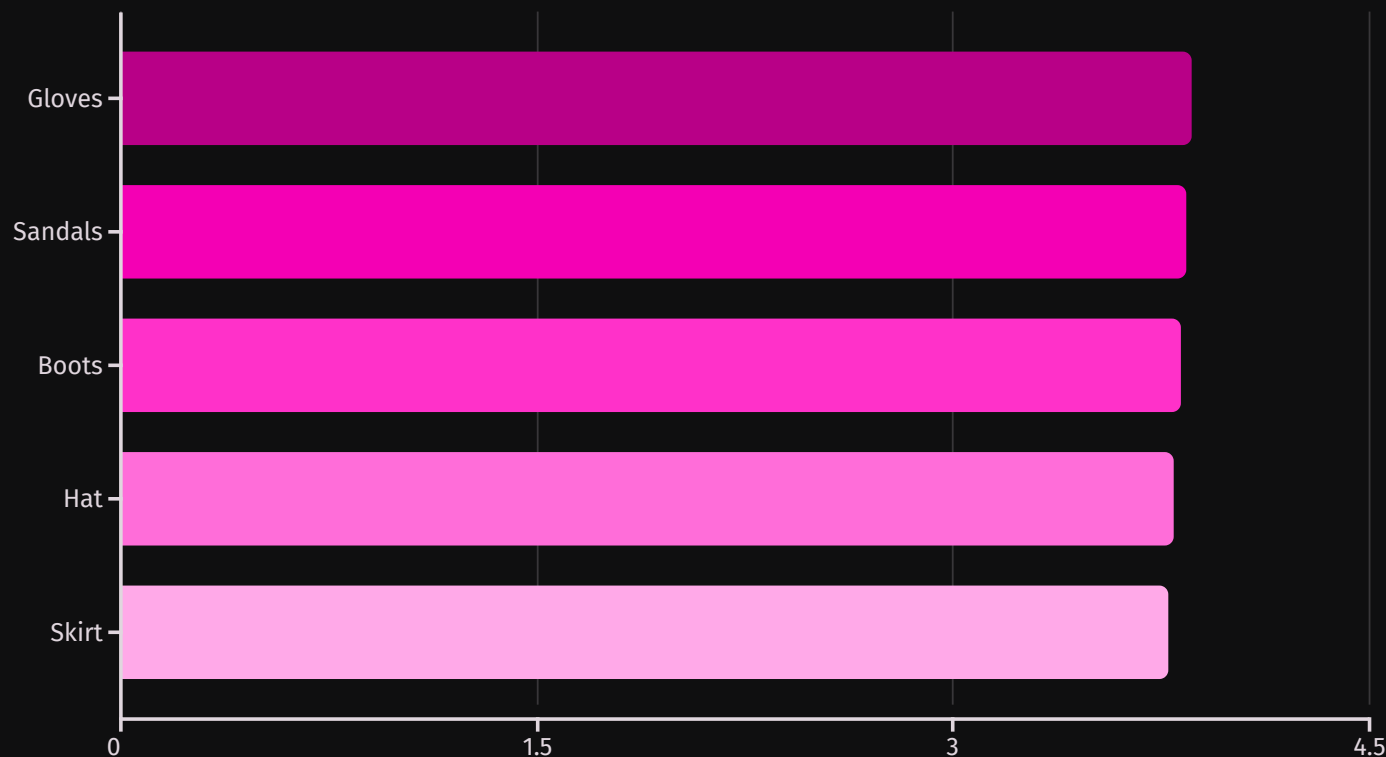
## Female Customers

\$75,191 total revenue

32% of total sales



# Top-Rated Products



## Customer Favorites

Gloves lead with 3.86 average rating, followed by Sandals and Boots

Accessories and footwear dominate top-rated products

# Customer Segmentation Insights



## Loyal Customers

3,116 customers (80%)

Strong retention base



## Returning Buyers

701 customers (18%)

Growth opportunity



## New Customers

83 customers (2%)

Acquisition focus needed

# Category & Age Group Performance

## Clothing

\$100K revenue

1,800 sales

## Accessories

\$70K revenue

1,200 sales

## Footwear

\$30K revenue

500 sales

## Outerwear

\$20K revenue

200 sales

## Age Group Leaders

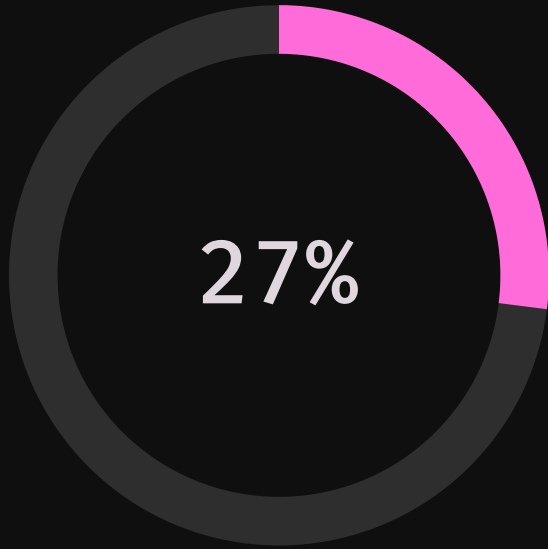
Young Adults: \$62K revenue

Middle-aged: \$59K revenue

Adults: \$56K revenue

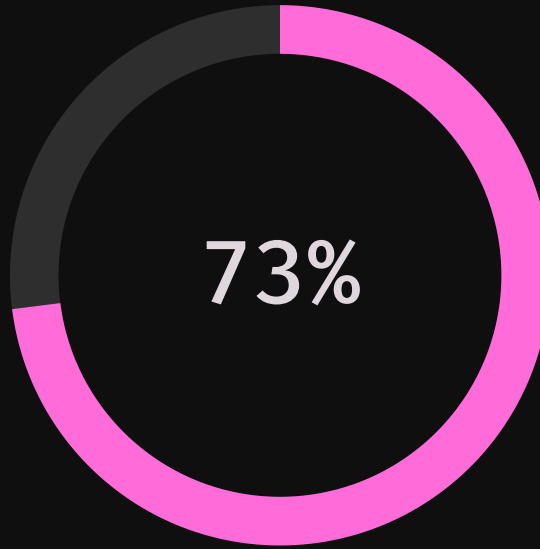
Seniors: \$56K revenue

# Subscription & Discount Behavior



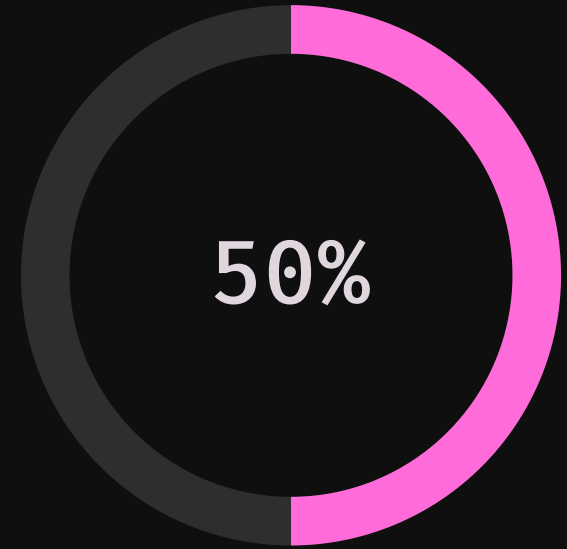
**Subscribers**

1,053 customers



**Non-Subscribers**

2,847 customers



**Discount Rate**

Hats most discounted

839 customers used discounts but spent above average—smart discount targeting opportunity



# Power BI Dashboard

## Real-Time Metrics

3.9K customers tracked

\$59.76 avg purchase

3.75 avg rating

## Visual Insights

Category performance

Age group analysis

Subscription trends

## Interactive Filters

Gender, category, shipping

Dynamic exploration



# Strategic Recommendations

01

## Boost Subscriptions

Promote exclusive benefits—only 27% currently subscribed

02

## Loyalty Programs

Move 701 returning buyers into loyal segment

03

## Optimize Discounts

Balance sales growth with margin control

04

## Product Positioning

Highlight top-rated items in marketing campaigns

05

## Targeted Marketing

Focus on Young Adults and express-shipping users